
MARGINAL MATCHING DOES NOT LICENSE FACTORIZED SAMPLING: AUDITING CONDITIONAL STYLE LEAKAGE IN FACTORIZED GENERATIVE MODELS

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ABSTRACT

Factorized generative models often regularize a style latent z_s toward a fixed prior using only the aggregate posterior $q(z_s)$, then sample z_s independently of the requested class. This marginal constraint does not certify the required conditional invariance $z_s \perp y$. We place the gap inside an exact decomposition of the latent mismatch induced by factorized sampling into global style-prior mismatch, conditional style leakage, semantic-prior mismatch, and within-class dependence between semantic and style codes. We then audit a class-structured WAE on MNIST, Fashion-MNIST, and CIFAR-10 using a fixed held-out protocol. Although stochastic style samples have global MMD^2 between 3.0×10^{-4} and 5.8×10^{-4} , a held-out logistic probe recovers the label with 99.76%, 88.29%, and $68.62 \pm 6.19\%$ accuracy, respectively (chance: 10%); permutation-calibrated HSIC rejects independence for every checkpoint and latent view ($p = 1/201$). Posterior means and samples give nearly identical results because the learned style posterior is almost deterministic, a failure mode we report rather than conceal. Finally, independent pixel-space classifiers show that replacing the global style prior with a post-hoc class-conditional prior raises generated-class accuracy from 0.1047 to 1.0000 on MNIST and from 0.1283 ± 0.0032 to 0.4716 ± 0.0686 on CIFAR-10. Marginal fit alone therefore neither verifies the claimed factorization nor validates the sampler built from it.

1 INTRODUCTION

Matching an aggregate posterior to a Gaussian is a statement about a marginal. Class-invariant style is a statement about conditionals. These claims are not interchangeable: $q(z_s)$ can equal $\mathcal{N}(0, I)$ while the conditionals $q(z_s | y = k)$ remain strongly separated. Any regularizer that sees only $q(z_s)$ is blind to how its mass is partitioned by y .

The distinction matters because factorized models are sampled differently from how they are encoded. During training, a decoder sees semantic and style codes from the same example. At generation, a class-conditional semantic prior is combined with an independently drawn global style code. If style is class-dependent, the decoder receives recombinations that need not follow the training joint. Reconstruction, clustering of z_c , and a global prior statistic can all look strong while this sampling contract fails.

This phenomenon is related to content leakage studied in earlier content–style models (Ridgeway & Mozer, 2018); our claim is not that label information in style was previously unknown. We instead ask what an independent factorized sampler must match, which parts aggregate objectives actually constrain, and how those distinctions should be audited. The answer is a four-term latent decomposition together with diagnostics whose latent view and evaluation split are stated explicitly.

Our empirical case study is F-CS-WAE (Factorized Class-Structured Spherical Cauchy WAE), a label-guided model with a hyperspherical semantic code and a Gaussian style code. We use it to expose a failure mode, not to claim a new state of the art. A new held-out audit evaluates both the deterministic posterior mean μ_s and stochastic sample z_s . This distinction is essential: the theory

concerns a random variable, whereas representation probes and deterministic swaps commonly use its mean.

Our contributions are:

1. an exact decomposition of factorized-sampler latent mismatch into four non-negative and operationally distinct terms;
2. a view-specific audit protocol using held-out probes, permutation-calibrated dependence tests, and decoder-level generation;
3. evidence on five checkpoints across three datasets that small global style MMD coexists with strong label information in stochastic style samples; and
4. a documented posterior-variance collapse that explains the agreement between mean- and sample-based diagnostics and delineates the present case study’s scope.

2 RELATED WORK

Aggregate matching and disentanglement. VAEs (Kingma & Welling, 2013) penalize a per-example KL, whereas WAEs (Tolstikhin et al., 2017) directly match aggregated posteriors using objectives such as MMD (Gretton et al., 2012). The averaged VAE KL decomposes into an information-capacity term and an aggregate-prior mismatch (Hoffman & Johnson, 2016); neither component alone certifies that a designated style code is independent of a particular label. β -VAE, β -TCVAE, FactorVAE, and DIP-VAE penalize capacity, total correlation, or aggregate moments (Higgins et al., 2017; Chen et al., 2018; Kim & Mnih, 2018; Kumar et al., 2018). These constraints differ from $I(z_s; y)$, and unsupervised disentanglement is not identifiable without assumptions on the model or data (Locatello et al., 2019).

Content–style separation and invariance. Split-latent models use grouped observations, labels, paired supervision, or latent optimization to separate specified and residual variation (Mathieu et al., 2016; Bouchacourt et al., 2018; Jha et al., 2018; Ilse et al., 2020; Gabbay & Hoshen, 2020). Most directly, Ridgeway & Mozer (2018) probe class information in style posteriors and introduce leakage filtering for recombination. Variational fair autoencoders, gradient reversal, Fader Networks, and information-based invariance objectives explicitly suppress designated nuisance information (Louizos et al., 2015; Ganin & Lempitsky, 2015; Lample et al., 2017; Moyer et al., 2018). Our decomposition clarifies why eliminating $I(z_s; y)$ is necessary but does not, by itself, make an independent sampler match the encoded joint.

Recombination. MUNIT and DRIT explicitly combine content with sampled style (Huang et al., 2018; Lee et al., 2018); more recent methods use variance patterns or invertible merging to obtain content–style structure (Wu et al., 2025; Ma et al., 2025). These works focus on learning useful separations. We focus on auditing the probabilistic claim that permits the two learned variables to be sampled independently.

3 FACTORIZED SAMPLING AND ITS MISMATCH

Let $q(y)$ denote the empirical label distribution and let an encoder induce $q(z_c, z_s | y)$. A factorized class-conditional sampler draws

$$z_c \sim p(z_c | y), \quad z_s \sim p(z_s), \quad (1)$$

independently. In our case $p(z_s) = \mathcal{N}(0, I)$. Many aggregate objectives constrain only

$$\mathcal{L}_{\text{style}} = D(q(z_s), p(z_s)), \quad q(z_s) = \mathbb{E}_{q(y)} q(z_s | y). \quad (2)$$

3.1 A MARGINAL CANNOT CERTIFY CONDITIONAL INVARIANCE

Proposition 1 (No marginal certificate). *Let $z_s \sim \mathcal{N}(0, I)$ and let $\{A_k\}_{k=1}^K$ be a measurable partition with $P(z_s \in A_k) = 1/K$. Define $y = k$ iff $z_s \in A_k$. Then $q(z_s) = \mathcal{N}(0, I)$ exactly for every marginal divergence D , while $I(z_s; y) = H(y) = \log K$.*

The construction is intentionally extremal, but the issue is not restricted to discontinuous partitions. For example, let $z \sim \mathcal{N}(0, 1)$ and $P(y = 1 | z) = \sigma(az)$. Symmetry gives $P(y = 1) = 1/2$ and leaves the marginal of z exactly Gaussian, while $q(z | y = 1) = 2\phi(z)\sigma(az)$ and $q(z | y = 0) = 2\phi(z)\sigma(-az)$ are smooth, full-support, and increasingly distinguishable as a grows. Thus the logical gap persists under regular conditional densities. Proofs are in Appendix A.

3.2 THE FOUR-TERM MISMATCH

Theorem 1 (Factorized-sampler latent mismatch). *For any $q(y)$, define*

$$M_{\text{fact}} := \mathbb{E}_{q(y)}[\text{KL}(q(z_c, z_s | y) || p(z_c | y)p(z_s))]. \quad (3)$$

Then

$$\begin{aligned} M_{\text{fact}} = & \underbrace{\text{KL}(q(z_s) || p(z_s))}_{(1) \text{ global style-prior mismatch}} + \underbrace{I_q(z_s; y)}_{(2) \text{ conditional style leakage}} \\ & + \underbrace{\mathbb{E}_{q(y)} \text{KL}(q(z_c | y) || p(z_c | y))}_{(3) \text{ semantic-prior mismatch}} + \underbrace{I_q(z_c; z_s | y)}_{(4) \text{ within-class dependence}}. \end{aligned} \quad (4)$$

All four terms are non-negative.

Corollary 1 (Exact latent sampling-contract match). $M_{\text{fact}} = 0$ iff $q(z_c, z_s | y) = p(z_c | y)p(z_s)$ for $q(y)$ -almost every y . Moreover, for a fixed decoder g ,

$$\mathbb{E}_{q(y)} \text{KL}(g_{\#} q(\cdot | y) || g_{\#} [p(z_c | y)p(z_s)]) \leq M_{\text{fact}}. \quad (5)$$

The inequality is one-way. A non-injective decoder can ignore mismatched latent directions, so latent equality is not necessary for identical decoded distributions; nor does equality between posterior-decoded and prior-decoded outputs establish agreement with real data. Decoder-level evaluation is therefore complementary to the decomposition.

Table 1 summarizes the operational consequence. The four terms answer different questions and cannot substitute for one another. In particular, a small value for term 1 says nothing about term 2, and removing term 2 still permits the decoder to exploit within-class dependence captured by term 4.

Table 1: The four latent requirements and their finite-sample audit. The statistics are proxies for distinct properties, not additive estimates of M_{fact} .

Term	Failure source	Required equality	Audit used here
1	Style-prior mismatch	$q(z_s) = p(z_s)$	global MMD
2	Conditional leakage	$q(z_s y) = q(z_s)$	probe, HSIC, Δ_{inter}
3	Semantic-prior mismatch	$q(z_c y) = p(z_c y)$	conditional MMD
4	Within-class coupling	$q(z_c, z_s y) = q(z_c y)q(z_s y)$	conditional HSIC

4 AUDIT PROTOCOL AND CASE STUDY

4.1 VIEW-SPECIFIC DIAGNOSTICS

We use stochastic posterior samples for claims about the sampling contract and means for deterministic representations:

- **Term 1:** unbiased multiscale-RBF MMD²($\{z_s^i\}, \mathcal{N}(0, I)$).
- **Term 2:** class-mean separation $\Delta_{\text{inter}} = \binom{K}{2}^{-1} \sum_{j < k} \|\bar{z}_s^{(j)} - \bar{z}_s^{(k)}\|_2$, a held-out probe suite, and a multiscale HSIC permutation test between style and label. We compute each on both μ_s and a seeded z_s draw.
- **Term 3:** class-conditional semantic MMD is part of training; we do not treat a minibatch training loss as a calibrated held-out estimate of the KL term.

- **Term 4:** classwise conditional HSIC with within-class permutations, computed on (μ_c, μ_s) and on ten seeded posterior draws (z_c, z_s) per checkpoint. It is a calibrated dependence test, not an estimate of conditional mutual information.

No finite-sample proxy is presented as an estimate of M_{fact} itself.

The fixed audit uses 2,048 test examples and a stratified 60/20/20 probe-train/validation/test split. Feature standardization is fit on probe training data only. We report logistic regression, a fixed two-layer MLP, RBF-SVM, and k -NN; hyperparameters and per-seed values are in Appendix C. Label HSIC uses 1,024 examples; conditional HSIC uses all 2,048. Both use 200 plus-one calibrated permutations, so the smallest attainable p -value is $1/201$. Global MMD uses an independent Gaussian RNG stream and a separate 500-permutation equality calibration.

4.2 CASE-STUDY MODEL

F-CS-WAE uses a shared modified ResNet-18 encoder, a hyperspherical semantic code $z_c \in \mathbb{S}^{63}$, a Gaussian style code $z_s \in \mathbb{R}^{128}$, and a residual upsampling decoder. The encoder trunk feeds separate heads for the semantic direction and concentration (μ_c, ρ_c) and style parameters $(\mu_s, \log \sigma_s^2)$. A forward pass samples both codes and decodes their concatenation. Class-specific Spherical Cauchy priors with EMA-updated centers shape z_c ; global MMD matches sampled z_s to $\mathcal{N}(0, I)$.

The complete objective is

$$\begin{aligned} \mathcal{L} = \mathcal{L}_{\text{rec}} + \alpha(t)\mathcal{L}_{\text{class}} + \beta(t)\mathcal{L}_{\text{agg}} + \gamma(t)\mathcal{L}_{\text{style}} \\ + \delta(t)\mathcal{L}_{\text{style-cl}} + \eta(t)\mathcal{L}_{\text{cls}}. \end{aligned} \quad (6)$$

$\mathcal{L}_{\text{rec}}$ combines L1 and LPIPS; the supervised class MMD matches $q(z_c | y = k)$ to its class prior, aggregate semantic MMD matches $q(z_c)$ to the prior mixture, global style MMD implements Eq. 2, and an auxiliary classifier on μ_c prevents semantic collapse. Training lasts 300 epochs in four phases: reconstruction warm-up, style regularization and classification, partial semantic matching, and the full objective. All coefficients and phase boundaries are listed in Appendix B.

The optional per-class loss

$$\mathcal{L}_{\text{style-cl}} = \frac{1}{K} \sum_k \text{MMD}^2(q(z_s | y = k), \mathcal{N}(0, I)) \quad (7)$$

targets term 2 but not term 4. It reuses labels already consumed by the semantic objective and is therefore a supervised intervention, not a generic unsupervised disentanglement loss. The held-out Stage-0 audit below focuses on unrepaired $\delta = 0$ checkpoints to isolate the failure of marginal matching. We subsequently use existing $\delta > 0$ checkpoints only to test whether explicitly conditional regularization moves the conditional distribution statistics in the predicted direction.

5 EXPERIMENTS

5.1 DATASETS, CHECKPOINTS, AND EVALUATION

We use MNIST (LeCun et al., 1998), Fashion-MNIST (Xiao et al., 2017), and CIFAR-10 (Krizhevsky, 2009). MNIST and Fashion-MNIST remain at their native 28×28 resolution; CIFAR-10 remains 32×32 . We use the standard train/test splits, scale pixels to $[0, 1]$, and apply no augmentation to the generative models. Stage-0 uses MNIST and Fashion-MNIST model seed 0 and CIFAR-10 seeds 0, 1, 2. All reported uncertainties are sample standard deviations over model seeds, not probe reruns. Each held-out probe test partition contains 410 examples.

We evaluate at three levels. Latent diagnostics use the fixed protocol of Section 4. Reconstruction uses SSIM and LPIPS (Zhang et al., 2018); clustering runs K -means on μ_c and reports Hungarian-matched ACC and NMI. Generation uses FID (Heusel et al., 2017) and generated-class accuracy (Gen-ACC). The latter is scored directly from generated pixels by classifiers trained only on real data: a four-layer CNN for MNIST (99.34% real-test accuracy) and a WideResNet-28-10 for CIFAR-10 (96.16%) (Zagoruyko & Komodakis, 2016). These classifiers never participate in F-CS-WAE training.

5.2 THE CASE STUDY IS NOT REPRESENTATION-DEGENERATE

Our audit would be uninformative if the case-study model failed to encode classes or reconstruct images at all. Table 2 gives a sanity check on CIFAR-10. F-CS-WAE reaches 0.812 ± 0.005 clustering ACC and 0.655 ± 0.008 NMI across three seeds. The comparison rows share a modified ResNet-18 and total latent dimension 192 but are single-seed and not equally tuned; we use them only to establish that the audited representation is nontrivial, not to claim controlled superiority.

Table 2: CIFAR-10 competence check. F-CS-WAE is mean $\pm$ sample standard deviation over three seeds; reference autoencoders are single-seed. Lower FID is better.

Method	ACC	NMI	FID
F-CS-WAE	$.812 \pm .005$	$.655 \pm .008$	81.9 ± 1.5
AE + cross entropy	.665	.611	144.3
AE + supervised contrastive	.541	.599	275.5
AE + triplet loss	.506	.502	281.4
ResNet autoencoder	.220	.102	121.0

5.3 SMALL GLOBAL MMD COEXISTS WITH LABEL LEAKAGE

Table 3: Held-out audit of unrepaired $\delta = 0$ checkpoints. MMD and Δ_{inter} use stochastic z_s ; LP columns are test accuracy. CIFAR-10 is mean $\pm$ sample standard deviation over three model seeds; the other rows are single checkpoints. Chance LP is 10%.

Dataset	MMD ²	Δ_{inter}	Logit μ_s	Logit z_s	RBF z_s
MNIST	.000579	6.264	99.76	99.76	99.76
Fashion-MNIST	.000336	4.791	88.29	88.29	93.17
CIFAR-10	$.000298 \pm .000052$	$3.954 \pm .269$	68.46 ± 6.29	68.62 ± 6.19	70.16 ± 3.36

Table 3 gives the central result. Global MMD is small on all five checkpoints, yet every held-out probe is far above chance. Multiscale HSIC independently rejects $z_s \perp y$ for both μ_s and z_s at every checkpoint ($p = 0.00498$). The result is not specific to linear separation: the MLP and RBF-SVM reach 90.73% and 93.17% on Fashion-MNIST, and $70.57 \pm 3.50\%$ and $70.16 \pm 3.36\%$ on stochastic CIFAR-10 codes. Full probe results appear in Appendix D.

A 500-permutation calibration places the 95th percentile of the equal- distribution MMD null between 1.88×10^{-5} and 2.45×10^{-5} ; exact equality is rejected at every checkpoint ($p = 1/501$). Thus “small” denotes numerical magnitude relative to the conditional discrepancy, not statistical indistinguishability from the prior. Posterior draws and Gaussian references use independent RNG streams.

The distinction between mean and sample matters conceptually but barely changes these checkpoints empirically. Across the reported probe suite the largest mean-sample difference is 0.49 percentage points on Fashion-MNIST and 0.73 points on any CIFAR-10 seed. Therefore the leakage persists in the stochastic variable used by Theorem 1; it is not an artifact of probing only μ_s .

5.4 WHY MEANS AND SAMPLES AGREE: POSTERIOR-VARIANCE COLLAPSE

The style encoder uses $z_s = \mu_s + \exp(\frac{1}{2} \log \sigma_s^2) \epsilon$ with $\epsilon \sim \mathcal{N}(0, I)$ and clamps log variance to $[-10, 10]$. Direct checkpoint inspection finds 99.9966% of MNIST, 99.9989% of Fashion-MNIST, and 100% of CIFAR-10 log-variance coordinates at or below -10 before the effective clamp. Hence the typical posterior standard deviation is $e^{-5} = 0.006738$, tiny relative to between-class separation 3.95–6.26. The learned stochastic encoder is thus almost deterministic.

This observation narrows, rather than inflates, our claim. It explains why sample-based verification agrees with mean-based diagnostics for these models, but does not imply that the two views are interchangeable in other models. It also exposes a limitation of the aggregate diversity penalty: class-dependent posterior means can create large batch variance without maintaining per-example posterior entropy.

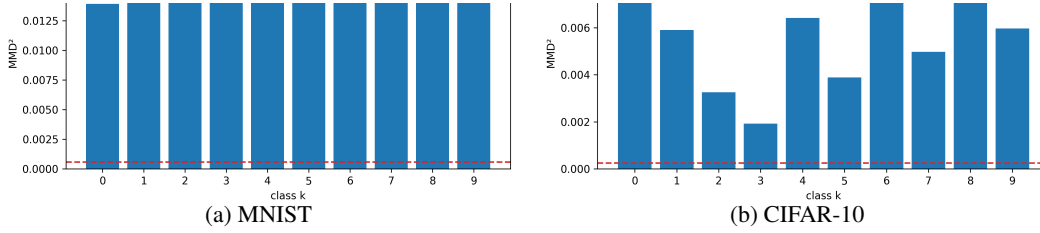


Figure 1: Per-class $\text{MMD}^2(q(z_s | y = k), \mathcal{N}(0, I))$ at $\delta = 0$ on stochastic posterior samples. The dashed line is the global $\text{MMD}^2(q(z_s), \mathcal{N}(0, I))$ optimized by the marginal regularizer. All ten conditional values exceed it on each dataset.

5.5 CONDITIONAL DISTRIBUTIONS REVEAL WHAT THE MARGINAL HIDES

The probe result is mirrored by a direct two-sample comparison. For each class we compute unbiased multiscale $\text{MMD}^2(q(z_s | y = k), \mathcal{N}(0, I))$ on the same 2,048-example subset and average over classes. Table 4 compares that value with the global MMD on seed-0 checkpoints. With only global style matching, mean conditional MMD is $27.7\times$ the global value on MNIST and $23.3\times$ on CIFAR-10. This is the distributional form of the same failure: mixture components remain class-specific even when their aggregate is close to the prior.

Table 4: Global versus mean per-class style MMD on stochastic z_s , single-seed checkpoints. The $\delta = 1$ model adds Eq. 7.

Dataset	δ	Global MMD	Cond. MMD	Ratio	Δ_{inter}
MNIST	0	.000579	.016024	27.68	6.264
MNIST	1	.000812	.001992	2.45	1.461
CIFAR-10	0	.000248	.005789	23.30	3.801
CIFAR-10	1	.000422	.000784	1.86	1.336

Figure 1 resolves the class average into its ten components. Every class-conditional MMD lies above the dashed global value on both datasets, so the gap is not driven by one or two exceptional classes. The marginal statistic is small precisely because it measures the mixture, not because its class-conditioned components individually match the prior.

The per-class objective acts in the predicted direction: it reduces mean conditional MMD by 87.6% on MNIST and 86.5% on CIFAR-10, and reduces Δ_{inter} by 76.7% and 64.9%, respectively. It does not make either conditional statistic vanish; global MMD remains below 10^{-3} . We do not reuse the historical remedy probe accuracy here because those checkpoints have not yet been regenerated with the held-out Stage-0 probe protocol.

The decomposition also predicts a distinct within-class failure. We sample (z_c, z_s) ten times per checkpoint and apply classwise conditional HSIC with within-class permutations. Every draw rejects $z_c \perp z_s | y$ on MNIST, Fashion-MNIST, and all three CIFAR-10 baseline seeds ($p = 1/201$). Rejection also persists for the seed-0 $\delta = 1$ checkpoints on MNIST and CIFAR-10. This is a stochastic-view kernel test, not an estimate of $I(z_c; z_s | y)$, but it directly demonstrates that targeting term 2 does not complete the term-4 audit.

5.6 CONDITIONAL REGULARIZATION HAS A NON-MONOTONE TRADE-OFF

Table 5 shows four points from the six-value δ sweep. Conditional separation decreases steadily, but clustering and FID are not monotone. On MNIST, the $\delta = 1$ checkpoint has an isolated ACC drop that disappears at $\delta = 3$; on CIFAR-10, ACC drops at $\delta = 0.3$ and only partially recovers. Because every sweep point is a single training seed, these irregularities may be optimization noise and are not evidence of a causal optimum.

Figure 2 makes the non-monotonicity visible across all six archived settings. The sweep supports a limited conclusion: conditional regularization consistently compresses class separation in style

Table 5: Selected points from the single-seed per-class-MMD sweep. No uncertainty or significance claim is made across settings.

δ	MNIST			CIFAR-10		
	ACC	Δ_{inter}	FID	ACC	Δ_{inter}	FID
0	.992	6.21	74.0	.813	3.74	80.5
.1	.990	2.12	52.6	.810	2.03	77.5
1	.859	1.22	43.4	.800	1.05	74.4
3	.993	1.18	41.5	.799	.95	76.1

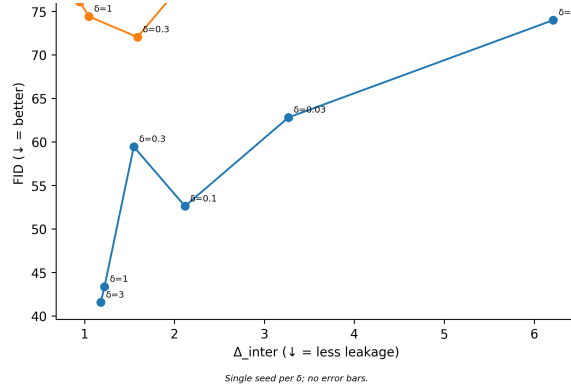


Figure 2: Naive-prior FID against Δ_{inter} across all six single-seed δ settings. Connecting lines are visual aids, not claims of a smooth response. Conditional separation decreases steadily, whereas FID improves overall but non-monotonically.

space, but reduced leakage does not guarantee monotonic improvement in semantic clustering or image quality. It also illustrates why the four terms of Theorem 1 should be monitored separately.

5.7 THE MISMATCH AFFECTS CLASS-CONDITIONAL GENERATION

We next test decoded samples using classifiers trained only on real pixels: a four-layer CNN with 99.34% MNIST test accuracy and a WideResNet-28-10 with 96.16% CIFAR-10 test accuracy (Zagoruyko & Komodakis, 2016). For a requested class k , the naive sampler draws $z_c \sim p(z_c | k)$ and $z_s \sim \mathcal{N}(0, I)$. A post-hoc alternative fits a diagonal Gaussian to encoded training styles from class k and samples with temperature $\tau = 0.25$. It uses labels and is a diagnostic workaround, not evidence that the representation is disentangled.

Table 6: External generated-class accuracy under global and post-hoc class-conditional style sampling. The post-hoc distribution is fitted on a training-only style bank; each checkpoint uses 1,000 generated images per requested class. CIFAR-10 is mean $\pm$ sample standard deviation over three model seeds.

Dataset	Global $\mathcal{N}(0, I)$	Class diagonal, $\tau = 0.25$
MNIST	0.1047	1.0000
CIFAR-10	0.1283 $\pm$ 0.0032	0.4716 $\pm$ 0.0686

Table 6 shows that respecting the learned class-conditional style distribution substantially improves class identity, especially on MNIST. The smaller CIFAR-10 gain is consistent with a diagonal conditional missing multimodal intra-class structure and with generally weak sample quality; it should not be read as a complete repair. A deterministic mean-swap intervention isolates decoder dependence. For off-diagonal donor pairs at $\delta = 0$, the external classifier follows the style donor in 99.28% of MNIST and 63.92% of CIFAR-10 decodes, versus the semantic donor in 0.06% and 4.72%, respectively. Per-class MMD reduces style-following to 23.37% and 35.92%, while increasing semantic-following; these remedy comparisons remain single-seed mechanism checks.

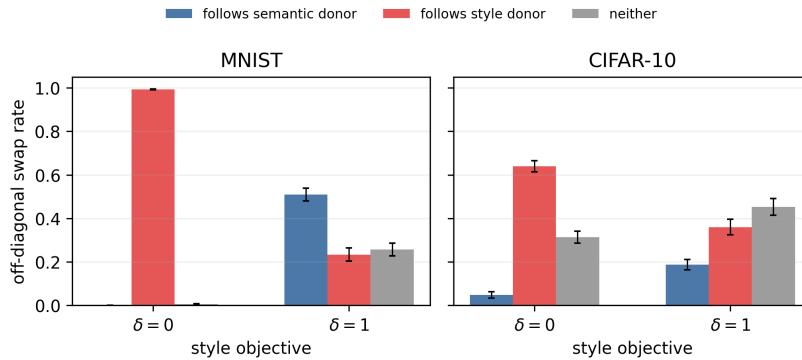


Figure 3: Independent-classifier outcomes for deterministic posterior-mean swaps. Bars average 90 ordered off-diagonal class pairs; intervals are 95% pair-bootstrap intervals. At $\delta = 0$, decoded identity predominantly follows the style donor; per-class style MMD reduces, but does not eliminate, this effect.

6 DISCUSSION AND LIMITATIONS

6.1 WHAT THE EVIDENCE CHAIN ESTABLISHES

The decomposition separates a logically necessary condition from an empirical failure. Proposition 1 shows that marginal matching cannot certify independence in any model. Table 3 shows the gap in actual stochastic posterior samples. Table 6 and Figure 3 then connect the discrepancy to decoded class identity through sampling and intervention.

Each step rules out a different, weaker explanation. The global-versus-conditional MMD comparison rules out interpreting a small mixture-level statistic as evidence that every class component matches the prior. Held-out linear and nonlinear probes, together with permutation-calibrated HSIC, show that the discrepancy is label-structured rather than merely an arbitrary distributional deviation. Repeating the audit on stochastic samples rules out a mean-only probing artifact for these checkpoints. Finally, the external generated-class evaluator connects the latent discrepancy to the sampler’s observable output rather than relying on re-encoded samples. No single diagnostic makes all four points.

The conditional-MMD intervention is supportive but deliberately secondary. It moves conditional MMD and class-mean separation in the direction predicted by term 2, while leaving global MMD almost unchanged. However, its sweep is single-seed, its quality metrics are non-monotone, and its post-hoc sampling counterpart uses labels. We therefore treat it as a mechanism check, not as a new disentanglement method or a tuned remedy benchmark.

6.2 IMPLICATIONS FOR EVALUATING FACTORIZED SAMPLERS

Theorem 1 suggests a minimal reporting standard. First, state the joint distribution that the decoder receives during encoding and the distribution used at generation; otherwise the claimed factorization has no testable contract. Second, identify whether each diagnostic uses posterior means, posterior samples, or prior samples. Third, report both aggregate fit and a conditional dependence statistic on held-out data. Fourth, examine within-class dependence between the proposed factors instead of stopping once style is label-invariant. Fifth, evaluate the final decoded samples with an external endpoint that was not trained jointly with the generator. The goal is not to estimate all four KL terms, but to align each proxy with the equality it tests.

6.3 LIMITATIONS AND THREATS TO VALIDITY

Several limitations bound the present evidence. First, class labels are a proxy for semantic content, not a complete definition of content. Datasets with known pose, lighting, shape, or color factors are needed to test both directions of disentanglement. Second, MNIST and Fashion-MNIST currently

have one model seed, while CIFAR-10 has three. Third, the almost deterministic style posterior may amplify leakage and motivates a controlled posterior- entropy ablation. Fourth, global and conditional MMD are finite-sample kernel statistics, not KL or mutual-information estimates. Fifth, generated class accuracy depends on an external classifier and does not measure sample quality or diversity. Finally, Eq. 7 is supervised and only targets term 2; even perfect conditional matching would leave terms 3 and 4 unresolved.

There are also finite-sample and model-selection threats. MMD and HSIC depend on kernel bandwidths and have limited power in high dimension; our multiscale kernels and permutation calibration reduce sensitivity to a single bandwidth but do not remove it. The probe split prevents test-set fitting, yet five checkpoints are insufficient to characterize training variability outside CIFAR-10. Generated-class accuracy can be biased by the external classifier’s decision boundary, so it should be paired with precision–recall or an equivalent diversity-sensitive measure in a definitive study. The present results are strongest as a falsification of the independence claim for these checkpoints, not as an estimate of how often the failure occurs across model families.

Our current scope is therefore an audit of a concrete factorized model, not a claim that all architectures fail at the same magnitude. The practical recommendation is correspondingly narrow: when independent style sampling requires $z_s \perp y$, report a conditional dependence test on stochastic samples in addition to aggregate prior fit, and verify whether the decoder uses the detected information.

7 CONCLUSION

Aggregate style matching does not license independent factorized sampling. In F-CS-WAE, held-out probes and permutation tests expose class information in stochastic style samples despite small global MMD; a conditional style sampler substantially improves generated-class accuracy. Separating marginal fit, conditional leakage, latent view, and decoder behavior turns prior fit into a falsifiable sampling audit.

REPRODUCIBILITY STATEMENT

Section 4 specifies the evaluation split, latent views, preprocessing, probe selection, and permutation calibration used for the headline audit. Appendix A contains complete proofs; Appendix B gives the architecture, objective coefficients, and training schedule; and Appendices C–E provide the canonical protocol, per-seed results, posterior-variance audit, and external generation evaluation. Appendix F records the checkpoint and artifact boundary for every reported claim. The evaluation code, configurations, and result manifests will be provided as an anonymous supplementary archive after a final identity and metadata audit.

ETHICS STATEMENT

This work uses standard public computer-vision benchmarks and existing model checkpoints, collects no new data, and involves no human subjects or sensitive personal information. Its purpose is to identify and measure a failure mode in factorized generative models. Generated examples are low-resolution benchmark images; nevertheless, the broader recommendation—auditing whether a decoder uses unintended latent information—may also help identify representation leakage in higher-stakes applications.

AI USE STATEMENT

Generative AI tools were used to provide feedback on evaluation methodology and experimental design, assist with interpretation of diagnostic results, audit consistency between implementation artifacts and reported procedures, and help restructure and edit the manuscript. They were not used to generate datasets or experimental measurements. The authors inspected the cited literature, verified the mathematical arguments, executed and tested the code, checked the experimental artifacts and every reported numerical value, and reviewed all AI-assisted text. The authors take responsibility for the final content of this work.

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A PROOFS

A.1 PROOF OF PROPOSITION 1

The label is defined after drawing z_s and therefore does not alter the marginal law of z_s . Hence $q(z_s) = \mathcal{N}(0, I)$ and any divergence between these two identical measures is zero. Because y is a deterministic function of z_s , $H(y | z_s) = 0$. The equal-probability partition gives $H(y) = \log K$, and consequently $I(z_s; y) = H(y) - H(y | z_s) = \log K$.

A.2 A SMOOTH, FULL-SUPPORT CONSTRUCTION

Let $z \sim \mathcal{N}(0, 1)$ and, conditional on z , draw a binary label with

$$P(y = 1 | z) = \sigma(az), \quad P(y = 0 | z) = \sigma(-az), \quad (8)$$

where $a > 0$ and $\sigma(t) = (1 + e^{-t})^{-1}$. Since the standard normal density ϕ is symmetric and $\sigma(t) + \sigma(-t) = 1$,

$$P(y = 1) = \int_{\mathbb{R}} \phi(z) \sigma(az) dz \quad (9)$$

$$= \frac{1}{2} \int_{\mathbb{R}} \phi(z) [\sigma(az) + \sigma(-az)] dz = \frac{1}{2}. \quad (10)$$

The generative construction starts with the standard normal and changes only the conditional label law, so the marginal of z remains exactly standard normal. Bayes' rule gives

$$q(z | y = 1) = 2\phi(z)\sigma(az), \quad q(z | y = 0) = 2\phi(z)\sigma(-az). \quad (11)$$

Both densities are smooth and strictly positive on $\mathbb{R}$. When $a \rightarrow \infty$, the conditional label approaches the indicator of the sign of z outside a set of Gaussian measure zero. Thus $H(y | z) \rightarrow 0$ and $I(z; y) \rightarrow \log 2$ while the marginal remains unchanged.

A.3 PROOF OF THEOREM 1

For a fixed y , insert the product of the encoder conditionals and apply the KL chain rule:

$$\text{KL}(q(z_c, z_s | y) \| p(z_c | y)p(z_s)) \quad (12)$$

$$= \text{KL}(q(z_c, z_s | y) \| q(z_c | y)q(z_s | y)) \\ + \text{KL}(q(z_c | y) \| p(z_c | y)) + \text{KL}(q(z_s | y) \| p(z_s)). \quad (13)$$

The first term is $I_q(z_c; z_s | y)$ at that label. Averaging over $q(y)$ gives term 4 and term 3 of Eq. 4. For the last term, insert the aggregate posterior $q(z_s)$:

$$\mathbb{E}_{q(y)} \text{KL}(q(z_s | y) \| p(z_s)) = \mathbb{E}_{q(y)} \text{KL}(q(z_s | y) \| q(z_s)) \\ + \text{KL}(q(z_s) \| p(z_s)) \quad (14)$$

$$= I_q(z_s; y) + \text{KL}(q(z_s) \| p(z_s)). \quad (15)$$

This establishes the equality for arbitrary $q(y)$. Non-negativity follows from non-negativity of KL divergence and mutual information.

A.4 PROOF OF COROLLARY 1

An expectation of a non-negative KL is zero iff its integrand is zero for $q(y)$ -almost every y , which holds iff the two latent conditional measures are equal. The pushforward inequality follows by applying the data-processing inequality to the deterministic Markov kernel induced by the decoder for each y , then averaging over $q(y)$.

B ARCHITECTURE AND TRAINING

B.1 ARCHITECTURE

The shared encoder is a ResNet-18 adapted to small images: its first 7×7 , stride-2 convolution is replaced by a 3×3 , stride-1 convolution and max pooling is removed. The trunk output is flattened and mapped to 256 dimensions with SiLU. Separate heads produce an unnormalized semantic direction, a scalar semantic concentration, a style mean, and a style log variance. We use

$$\mu_c = \widetilde{\mu}_c / \|\widetilde{\mu}_c\|_2, \quad \rho_c = \sigma(r_c)(1 - \epsilon), \quad (16)$$

$$z_s = \mu_s + \exp(\frac{1}{2} \log \sigma_s^2) \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I). \quad (17)$$

The semantic code is sampled by a Spherical Cauchy Möbius reparameterization. The decoder concatenates $[z_c; z_s]$, projects it to a $512 \times 4 \times 4$ tensor, and applies three residual bilinear-upsample blocks with channel widths 256, 128, 64. Its sigmoid output is interpolated to the native dataset resolution when necessary. MNIST and Fashion-MNIST therefore remain 28×28 and single-channel; CIFAR-10 remains $32 \times 32 \times 3$.

For class k , the semantic prior is a Spherical Cauchy distribution centered at an EMA direction m_k , with concentration $\rho_p = 0.7$. The center is updated once per epoch from encoded class-mean directions with momentum 0.95. The style prior is $\mathcal{N}(0, I)$.

B.2 OBJECTIVE

The training objective is

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \alpha(t)\mathcal{L}_{\text{class}} + \beta(t)\mathcal{L}_{\text{agg}} + \gamma(t)\mathcal{L}_{\text{style}} + \delta(t)\mathcal{L}_{\text{style-clas}} + \eta(t)\mathcal{L}_{\text{cls}}. \quad (18)$$

$\mathcal{L}_{\text{rec}}$ is L1 plus 0.1 LPIPS (Zhang et al., 2018). $\mathcal{L}_{\text{class}}$ matches each semantic class conditional to its spherical prior, $\mathcal{L}_{\text{agg}}$ matches aggregate semantic codes to the class-prior mixture, $\mathcal{L}_{\text{style}}$ matches sampled style codes to a standard Gaussian, and $\mathcal{L}_{\text{cls}}$ is cross entropy from μ_c . The baseline checkpoints audited in the main paper explicitly set $\delta = 0$; the repository default for new remedied runs is $\delta = 1$, so the run manifest or command-line override is required to distinguish them.

Table 7: Case-study training hyperparameters.

Parameter	Value
Optimizer	Adam, learning rate 10^{-3}
Schedule	StepLR, step 100, factor 0.5
Batch size / epochs	128 / 300
Gradient clipping	global norm 1.0
Semantic / style dimension	64 / 128
Semantic-prior concentration	0.7
EMA momentum	0.95
Final class / aggregate MMD weights	2.0 / 5.0
Final global style MMD weight	1.0
Final auxiliary-classifier weight	0.3

Training has four phases: reconstruction only for epochs 0–49; global style MMD and the auxiliary classifier ramp during epochs 50–99; semantic class and aggregate MMD ramp to half strength during epochs 100–199; all terms reach full strength during epochs 200–299. For remedied runs, per-class style MMD follows the global-style ramp. No image augmentation is used.

C CANONICAL AUDIT PROTOCOL

C.1 DATA SPLIT AND LATENT VIEWS

For each checkpoint we select 2,048 examples from the standard test split with evaluation seed 0. Probe indices are generated by a deterministic stratified 60/20/20 split, yielding 1,228 training, 410 validation, and 410 test examples. Standardization parameters are estimated only on the 1,228 probe-training examples. The split indices are stored by SHA-256 hash in each result manifest.

The encoder is evaluated once to obtain $(\mu_c, \mu_s, \log \sigma_s^2)$ and a seeded stochastic draw z_s . We use:

- μ_s for representation probes and deterministic mean swaps;
- z_s for global MMD, conditional style statistics, and the sampling-contract probe; and
- both (μ_c, μ_s) and repeated posterior draws (z_c, z_s) for the conditional-dependence diagnostic.

Agreement between these views is reported as an empirical result rather than assumed.

C.2 PROBE SUITE

The suite contains multinomial logistic regression, a fixed two-layer MLP, RBF-SVM, and k -NN. Validation data are used for fixed early-stopping or model-selection rules; every headline accuracy is evaluated once on the untouched test partition. We report accuracy and macro-F1. The main paper uses logistic regression as the simplest standardized headline and includes RBF-SVM to detect nonlinear separation.

C.3 MMD AND HSIC

Global MMD is the unbiased U-statistic with the averaged Euclidean RBF kernel

$$k(u, v) = \frac{1}{7} \sum_{\sigma \in \{0.5, 1, 2, 5, 10, 20, 50\}} \exp\left(-\frac{\|u - v\|_2^2}{2\sigma^2}\right). \quad (19)$$

The Gaussian reference has the same sample count as the encoded set and uses an RNG stream independent of posterior sampling. A balanced pooled-label permutation test calibrates the finite-sample null with 500 permutations. The resulting null 95th percentiles are $1.88\text{--}2.45 \times 10^{-5}$, whereas observed global MMD is $2.48\text{--}5.79 \times 10^{-4}$; exact equality is rejected for every checkpoint ($p = 1/501$). We therefore use “small” only as a relative numerical description, not as acceptance of the equality null. Δ_{inter} is not an MMD; it is the average Euclidean distance between all pairs of class-conditional style means.

HSIC centers a multiscale style Gram matrix and a label Gram matrix and uses 1,024 examples for bounded memory. We calibrate it by 200 label permutations. With the plus-one correction,

$$p = \frac{1 + \#\{T_b \geq T_{\text{obs}}\}}{201}, \quad (20)$$

so $p = 0.004975$ is reported rather than $p = 0$. Conditional HSIC for term 4 permutes styles independently inside every represented class.

We do not report the current JointMMD U-statistic as a headline result. Its joint and recombined samples share observations, so the nominal independent two-sample U-statistic assumptions require a sample-splitting calibration that is not yet part of the canonical protocol.

D FULL STAGE-0 RESULTS

Table 8: Full held-out probe suite. Values are test accuracy in percent. CIFAR-10 is mean $\pm$ sample standard deviation over model seeds 0, 1, 2.

Dataset	View	Logistic	MLP	RBF-SVM	k-NN
MNIST	μ_s	99.76	99.51	99.76	97.80
	z_s	99.76	99.51	99.76	97.80
Fashion	μ_s	88.29	90.73	93.17	85.12
	z_s	88.29	90.73	93.17	85.61
CIFAR-10	μ_s	68.46 ± 6.29	70.49 ± 3.83	70.24 ± 3.23	42.52 ± 6.93
	z_s	68.62 ± 6.19	70.57 ± 3.50	70.16 ± 3.36	42.20 ± 7.11

Table 9: Per-seed CIFAR-10 results for the two principal probes.

Seed	$\Delta_{\text{inter}}(z_s)$	Logit μ_s	Logit z_s	RBF μ_s	RBF z_s
0	3.801	61.46	61.71	66.59	66.34
1	4.264	73.66	73.66	72.68	72.68
2	3.797	70.24	70.49	71.46	71.46

For term 4, ten stochastic posterior draws per checkpoint all attain the minimum calibrated $p = 1/201$. The sample-view statistic is 0.002682 ± 0.000005 across draws on MNIST, 0.002459 on Fashion-MNIST, and has per-model means 0.002250 , 0.002292 , and 0.002338 on CIFAR-10. The seed-0 $\delta = 1$ checkpoints also reject on all ten draws, with means 0.002507 (MNIST) and 0.002283 (CIFAR-10). These calibrated kernel tests are evidence of stochastic within-class dependence, not numerical estimates of $I(z_c; z_s | y)$.

D.1 STRUCTURE OF THE CONDITIONAL MISMATCH

The class average in Table 4 compresses a structured ten-class discrepancy. Figure 4 compares every pair of class-conditional style distributions. Broad off-diagonal mass is visible on both datasets,

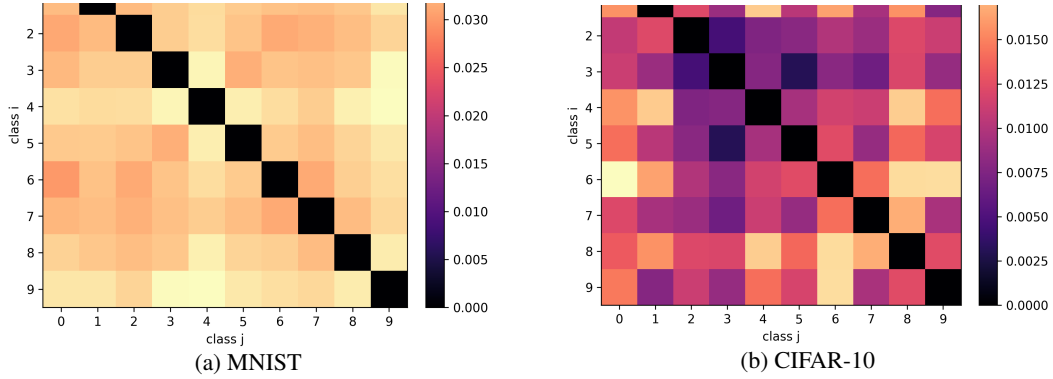


Figure 4: Pairwise $\text{MMD}^2(q(z_s | y = i), q(z_s | y = j))$ for the unrepaired seed-0 checkpoints. The diagonal is zero by construction; off-diagonal structure shows that conditional style distributions differ throughout the label set.

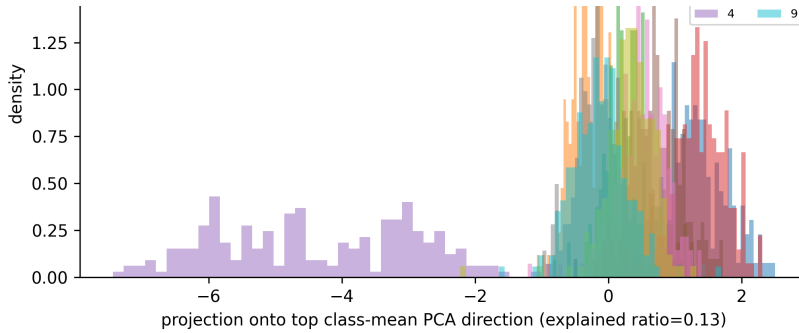


Figure 5: MNIST stochastic styles projected onto the leading direction of the class-mean matrix at $\delta = 0$. The direction explains 13.1% of the class-mean variance. The plot is descriptive; the held-out probes and HSIC tests provide the quantitative evidence.

rather than separation being confined to a few class pairs. These matrices use the same stochastic-view diagnostic as the per-class prior comparison; their scale should not be compared across datasets as if it were a normalized dependence measure.

Figure 5 gives a complementary one-dimensional view. The projection direction is computed from the class means rather than chosen after inspecting individual examples. Several MNIST classes separate along this direction, but the overlap also illustrates why a single projection is not a substitute for the held-out multivariate probes.

D.2 POSTERIOR-VARIANCE INSPECTION

The fractions of raw style log-variance coordinates at or below the clamp floor are 99.9966% (MNIST), 99.9989% (Fashion-MNIST), and 100% for all three CIFAR-10 checkpoints. Because this statistic was added during the post-hoc checkpoint audit, it is reported in the human-readable audit report; future manifests should also store it as a first-class numeric field.

D.3 COMPLETE CONDITIONAL-REGULARIZATION SWEEP

Table 10 expands the four points shown in the main paper to all six archived δ settings. We retain ACC, Δ_{inter} , and FID because they are directly tied to the sweep artifacts, but omit the historical probe column until it is regenerated with the canonical held-out split. Each point is a separate single-seed training run.

Table 10: Complete single-seed per-class-style-MMD sweep. Lower FID and Δ_{inter} are better; no uncertainty or significance claim is made.

δ	MNIST			CIFAR-10		
	ACC	Δ_{inter}	FID	ACC	Δ_{inter}	FID
0	.9915	6.21	74.0	.8129	3.74	80.5
.03	.9944	3.27	62.8	.8111	2.48	79.1
.1	.9898	2.12	52.6	.8100	2.03	77.5
.3	.9936	1.55	59.4	.7148	1.59	72.0
1	.8592	1.22	43.4	.8000	1.05	74.4
3	.9929	1.18	41.5	.7994	.95	76.1

E GENERATION AND DECODER DIAGNOSTICS

E.1 EXTERNAL GENERATION EVALUATION

The independent MNIST evaluator is a four-convolution CNN trained for 20 epochs on real images only. Its standard test accuracy is 99.34%. The CIFAR-10 evaluator is a WideResNet-28-10 trained for 200 epochs with standard random-crop and horizontal-flip augmentation on real training images only; its test accuracy is 96.16%. Neither evaluator contributes to the generative model’s loss.

Table 11: Per-checkpoint external Gen-ACC. Each entry uses 1,000 images per requested class and a class-conditional style bank fitted on 12,000 training examples.

Dataset	Model seed	Global Gaussian	Class diagonal $\tau = .25$
MNIST	0	0.1047	1.0000
CIFAR-10	0	0.1262	0.4016
CIFAR-10	1	0.1320	0.5386
CIFAR-10	2	0.1268	0.4747

The class-diagonal distribution is fit from encoded training styles and therefore deliberately conditions style on the requested label. Its gain tests whether the decoder is sensitive to the conditional mismatch. It is not an unconditional generative model and does not repair the encoder.

Figure 6 displays the qualitative counterpart of Table 11. Conditioning style makes the requested category more visually consistent in some rows, but shrinking the diagonal Gaussian to $\tau = .25$ also reduces variation and can produce blurry or prototype-like outputs. This is why Gen-ACC is not interpreted as a stand-alone measure of generative quality; diversity-sensitive evaluation on the same generated pools remains necessary.



Figure 6: Archived CIFAR-10 samples from the seed-0 checkpoint. For four requested classes, odd rows use the global Gaussian style prior and even rows use the post-hoc class-diagonal prior at $\tau = .25$. The grid is qualitative; external Gen-ACC is reported in Table 11.

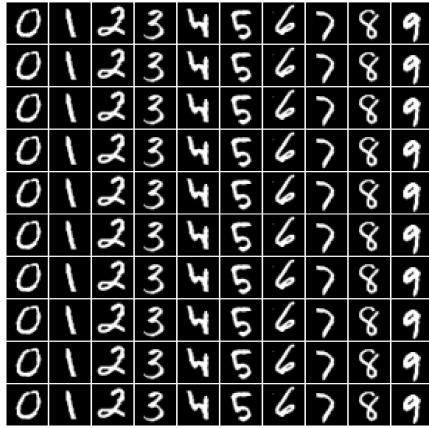
E.2 INDEPENDENT-EVALUATOR DETERMINISTIC MEAN SWAPS

For each ordered pair of distinct classes (a, b) , the diagnostic decodes $\text{Dec}(\mu_c^{(a)}, \mu_s^{(b)})$ for 500 donor pairs and evaluates pixels with the independent real-image classifier. On unrepaired checkpoints, identity follows the style donor on 99.28% of MNIST and 63.92% of CIFAR-10 off-diagonal swaps, while following the semantic donor on 0.06% and 4.72%. At $\delta = 1$, style-following falls to 23.37% and 35.92%, and semantic- following rises to 51.00% and 18.74%, respectively. Full rate matrices, the other-class outcome, and pair-bootstrap intervals are stored in the external-swap artifacts.

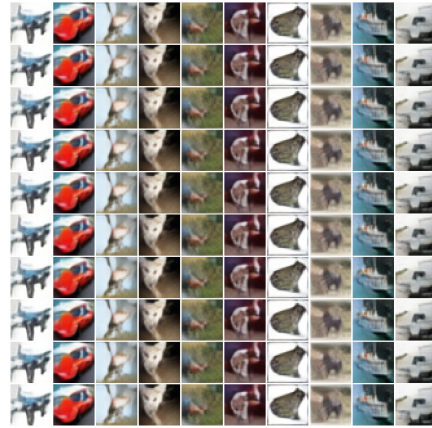
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Latent swap: row = z_c class (content), col = z_s class (style) — mnist_delta0

Latent swap: row = z_c class (content), col = z_s class (style) — cifar10_delta0



(a) MNIST



(b) CIFAR-10

Figure 7: Deterministic mean-swap grids at $\delta = 0$. Rows donate semantic means and columns donate style means. Column-controlled identity is visually strong on MNIST and weaker but still present for several CIFAR-10 classes. Quantitative rates in the text use independent pixel classifiers; the grids are qualitative.

F REPRODUCIBILITY AND CURRENT EVIDENCE BOUNDARY

All Stage-0 results are stored as JSON plus SHA-256 sidecars. Each manifest records the repository state, checkpoint hash, evaluation seed, fixed subset, latent view, probe split, feature preprocessing, and evaluator metadata. The canonical protocol version for the present draft is `stage0-1.1.0`. The five audited checkpoints are MNIST seed 0, Fashion-MNIST seed 0, and CIFAR-10 seeds 0, 1, 2; all use evaluation seed 0.

The current draft intentionally excludes the earlier cross-model probe table, multi-remedy comparison, five-seed remedy/sampling tables, and explicit split-latent baseline ranges. Those measurements were not yet regenerated under the held-out Stage-0 protocol with complete artifacts. They should be reintroduced only after model-specific loaders, manifests, and canonical outputs are available. This boundary keeps every quantitative claim in the present paper tied to an inspectable artifact.

The submission source, rendered author block, supplementary archive, and all externally hosted materials must remain anonymous throughout review.